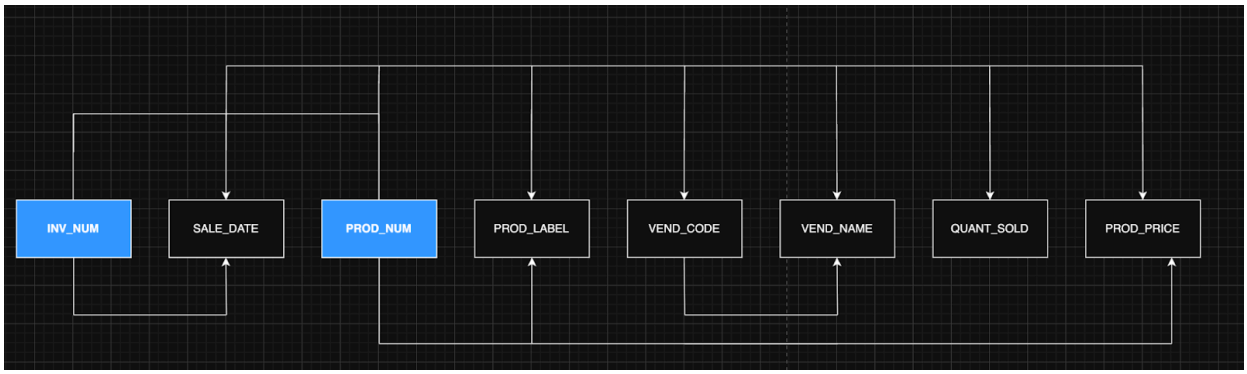
Relational Databases

1. Restructuring database tables to minimise redundancies and reduce data anomalies - bringing the database to a consistent state.
2. **A table is in 1NF when:**
    1. All of the key attributes are defined.
    2. There are no repeating groups.
    3. All attributes are dependent on the primary key.
3. **A table is in 2NF when:**
    1. It is in 1NF.
    2. It includes no partial dependencies - no attribute is dependent on only a portion of the primary key.
4. **A table is in 3NF when:**
    1. It is in 2NF.
    2. It contains no transitive dependencies.

5. 1NF:



6. 2NF:

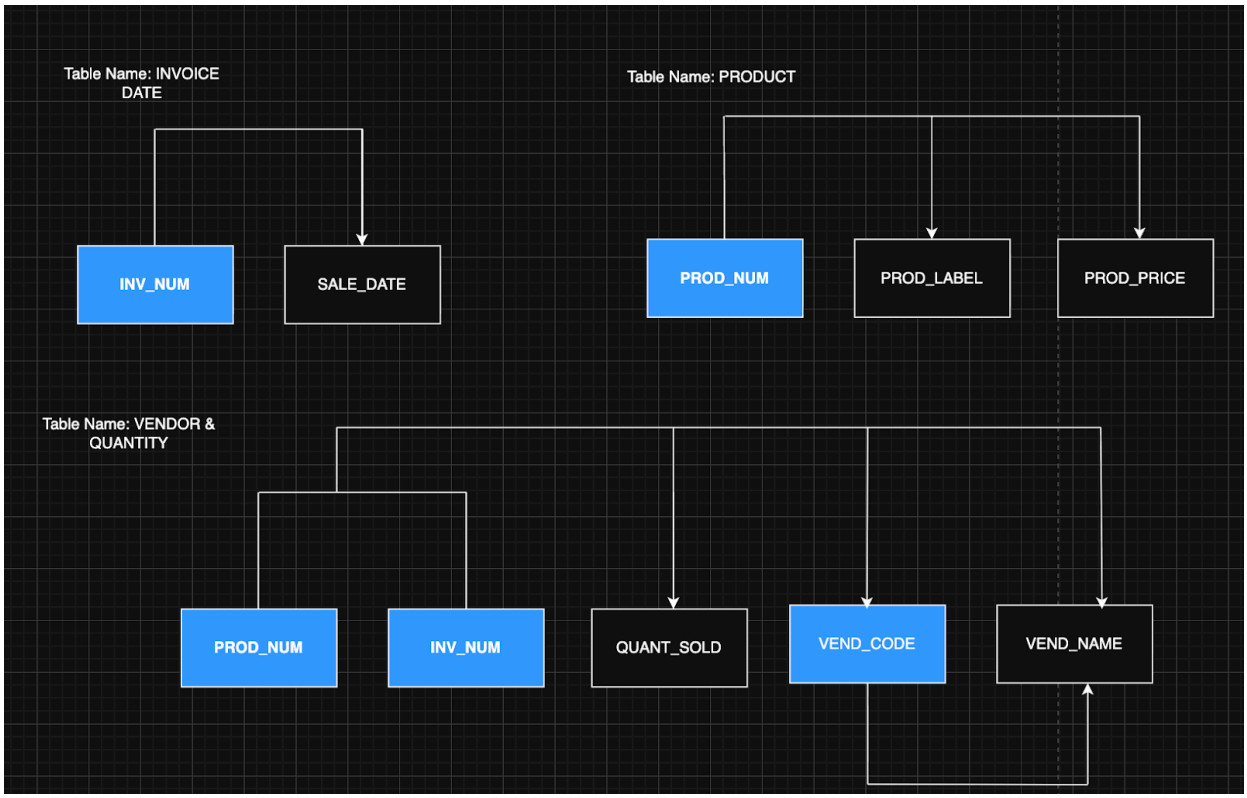


Table Name: INVOICE DATE
INV_NUM
SALE_DATE
Table Name: PRODUCT
PROD_NUM
PROD_LABEL
PROD_PRICE
Table Name: VENDOR & QUANTITY
PROD_NUM
INV_NUM
QUANT_SOLD
VEND_CODE
VEND_NAME

7. 3NF:

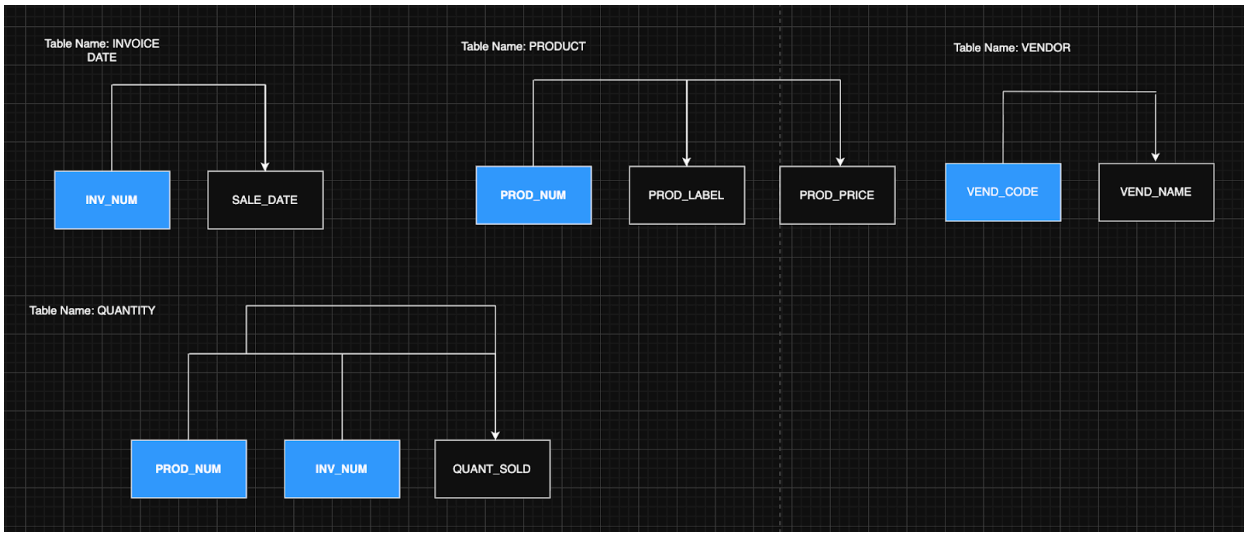


Table Name: INVOICE DATE
INV_NUM
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PROD_NUM
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VEND_CODE
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PROD_NUM
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